Doron Haviv

Research Scientist at Genentech

doron.haviv12@gmail.com Doron Haviv linkedin.com/in/doron-haviv doronhav.github.io

EMPLOYMENT

Genentech BRAID - Biology Research — AI Development May 2025 – Present

Senior AI Scientist

EDUCATION

Cornell University Joint Memorial Sloan Kettering Cancer Center July 2019 – April 2025

Ph.D. in Computational Biology and Medicine

Research Advisor: Dana Pe'er
Thesis: From Cells to Niches - The Geometry of Spatial Transcriptomics

Technion - Israel Institute of Technology October 2014 – October 2018

B.Sc. Electrical Engineering (*cum laude*)

B.Sc. Physics (*cum laude*)

Research Advisor: Omri Barak
Thesis: Understanding and Controlling Memory in Recurrent Neural Networks.

RESEARCH INTERESTS

generative models, single-cell genomics, optimal transport, spatial transcriptomics

PUBLICATIONS AND PRE-PRINTS

Lee, J., Scalia, G., BenTaieb, A., Haviv, D.*, De Brouwer, E.* 2026. Solving Inverse Problems of Distributions with Wasserstein Flow Maps. **In review**.

Haviv, D.*, De Brouwer, E.*, Rishabh, A., Ying, R., Scalia, G., Corrada-Bravo, H. 2026. When Riemann flows with Wasserstein: Generative Modeling of Probability Distributions on Manifolds. **NeurIPS**.

Andersson, A.*, Ismail, A.A.*, De Brouwer, E.*, Haviv, D.*, Biancalani, T., Cho, K., Scalia, G., BenTaieb, A., Corrada-Bravo, H. 2026. scCBGM: Single-Cell Editing via Concept Bottlenecks. **ICML**.

Haviv, D.*, Pooladian, A.A.*, Pe'er, D., Amos, B. 2025. Wasserstein Flow Matching: Generative modeling over families of distributions. **ICML**.

Haviv, D., Kunes, R.Z., Dougherty, T. Burdziak, C., Nawy T., Gilbert A., Pe'er, D. 2024. Wasserstein Wormhole: Scalable Optimal Transport Distance with Transformers. **ICML**.

Haviv, D.*, Remšík, J., Gatie, M., Snopkowski, C., Takizawa, M., Pereira, N., ..., Pe'er, D. 2024. The covariance environment defines cellular niches for spatial inference. **Nature Biotechnology**.
Highlighted in *Nature Biotechnology Research Briefing

Mani, S.*, Haviv, D.*, Kunes, R., Pe'er, D. 2022. SPOT: Spatial Optimal Transport for Analyzing Cellular Microenvironments. **NeurIPS Workshop on Learning Meaningful Representations of Life**.
***Spotlight Presentation**

Elad, A.*, Haviv, D.*, Blau, Y. and Michaeli, T., 2019. Direct validation of the information bottleneck principle for deep nets. **ICCV Workshop on Statistical Deep Learning in Computer Vision**.
***Best poster award**

Haviv, D., Rivkind, A. and Barak, O., 2019. Understanding and controlling memory in recurrent neural networks. **ICML**.

Kunes, R.Z., Yin, M., Land, M., Haviv, D., Pe'er, D. and Tavaré, S., 2023. Gradient estimation for binary latent variables via gradient variance clipping. **AAAI**.

Burdziak, C.*, Zhao, C. J.*, Haviv, D., Alonso-Curbelo, D., Lowe, S. W., Pe'er, D. 2023. scKINETICS: inference of regulatory velocity with single-cell transcriptomics data. **Bioinformatics**.
***Best paper award Intelligent Systems for Molecular Biology (ISMB) 2023.**

Burdziak, C.*, Alonso-Curbelo, D.*, Walle, T., Reyes, J., Barriga, F. M., Haviv, D., Xie, Y., Zhao, Z., Zhao, C. J., Chen, H.-A., Chaudhary, O., Masilionis, I., Choo, Z.-N., Gao, V., Luan, W., Wuest, A., Ho, Y.-J., Wei, Y., Quail, D. F., ... Pe'er, D. 2023. Epigenetic plasticity cooperates with cell-cell interactions to direct pancreatic tumorigenesis. **Science**.
***Highlighted in *Cancer Discovery*, *Nature Reviews Gastroenterology and Hepatology*, *Cell Trends in Cancer*, and EACR Highlights in Cancer Research.**

Raayoni, G., Gottlieb, S., Manor, Y., Pisha, G., Harris, Y., Mendlovic, U., Haviv, D., Hadad, Y. and Kaminer, I., 2021. Generating conjectures on fundamental constants with the Ramanujan Machine. **Nature**.
Highlighted in *New Scientist

AWARDS AND HONORS

Best Poster Award, SDL-VC workshop, International Conference on Computer Vision 2019
Yehoraz Kasher Prize **Best Student Project in Electrical Engineering, 3rd Place** 2018
Technion - Israel Institute Of Technology **President's List** 2018
Technion - Israel Institute Of Technology **Dean's List** 2017
Technion - Israel Institute Of Technology **Dean's List** 2016

INVITED AND CONTRIBUTED TALKS

Learning to Match Distributions: Optimal Transport, Flow Matching & Applications class
NYU, New York, New York, USA 2026
Guest Lecture: *Learning flows of distributions of distributions*

Computational Genomics class, Columbia University, New York, New York, USA 2026
Guest Lecture: *Spatial Transcriptomics - Technologies and Algorithms*

Apple Machine Learning Research Group, Virtual 2024
Invited Talk: *Wasserstein Wormhole: Scalable Optimal Transport Distance with Transformers.*

scverse Community Meeting, Virtual 2024
Invited Talk: *Reconstructing spatial context for single cell transcriptomics with ENVI*

Department of Computer Science Colloquium, Columbia University, New York, New York, USA 2024
Invited Talk: *Reconstructing spatial context for single cell transcriptomics with ENVI*

Ido Amit Lab, Weizmann Institute of Science, Rehovot, Israel 2024
Invited Talk: *Reconstructing spatial context for single cell transcriptomics with ENVI*

10x Spatial World Tour, New York Genome Center, New York, New York, USA 2023
Invited Talk: *Reconstructing spatial context for single cell transcriptomics with ENVI*

The Jackson Laboratory for Genomic Medicine, Farmington, Connecticut, USA 2022
Invited Talk: *Reconstructing spatial context for single cell transcriptomics with ENVI*

Fusion Conference on Probing Human Disease using Single-Cell Technologies, Cancun, MX 2022
Contributed Talk: *Spatial Context of Heterogeneous T Cell Response to Fungal Insult.*

International conference on machine learning. Long Beach, California, USA 2019
Contributed Talk: *Understanding and controlling memory in recurrent neural networks*

TEACHING AND MENTORSHIP

Intern Mentor, Genentech

- **Jungyoon Lee** 2026
  Solving Inverse Problems of Distributions with Wasserstein Flow Maps
  Ph.D. student at MILA

Intern Mentor, Dana Pe'er Lab, Memorial Sloan Kettering Cancer Center
B.Sc./M.Sc. student interns

- **Emma Chen** 2024
  MINGLE: counterfactual inference of niches from spatial transcriptomics data.
  Now a Ph.D. student at Stanford University
- **Shouvik Mani** 2022
  Spatial Optimal Transport for analyzing cellular microenvironments
  Now a Ph.D. student at Stanford University
- **Yasa Baig** 2021
  Discrete latent models for interpretable single-cell analysis
  Now a Ph.D. student at Stanford University

Teaching Assistant, Technion - Israel Institute Of Technology, 2018-2019
Introduction to Biological Systems and Signals, Head TA
Electromagnetic Fields

REVIEWING

- **Journals**: Nature Biomedical Engineering, Nature Biotechnology, Nature Methods, Cell, Genome Biology
- **Conferences**: NeurIPS, ICLR, ICML, ICML Workshop in Computational Biology